

MICHAEL C. PUENTE

MECHANICAL ENGINEERING STUDENT

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mcpuenteBSME.github.io

Mechanical engineering student seeking internships in design, manufacturing, aerospace, or thermal-fluid systems. Combines computational analysis, CAD, commercial MEP experience, and hands-on troubleshooting.

EDUCATION

The University of Texas at San Antonio - B.S. Mechanical Engineering **Expected Fall 2027**

Coursework: Machine Element Design, Manufacturing Engineering, Measurements and Instrumentation, Numerical Methods, Mechanics of Solids, Dynamic Systems and Controls

Texas Lutheran University - Applied Physics coursework; varsity football

Activities: American Society of Mechanical Engineers; Society of Physics Students

RESEARCH AND INDEPENDENT STUDY

Hypersonic Turbulent Boundary Layers | UTSA **Ongoing**

- Developing a Mach 6 adiabatic flat-plate baseline in OpenFOAM using URANS to study boundary-layer growth, wall friction, and pressure changes; rectangular-cavity simulation is the next objective.
- Use ParaView and MATLAB to examine flow fields, estimate boundary-layer thickness and skin friction, and generate Mach-number, static-temperature, streamwise-velocity, and wall-unit plots.

Undergraduate Researcher | UTSA Klesse Program **Summer 2026**

- Set up CFD meshes and boundary conditions, ran parallel OpenFOAM jobs on TACC and ARC systems, and interpreted results in ParaView/ParaFoam.

ENGINEERING AND TECHNICAL EXPERIENCE

Mechanical Engineering Intern | Sur Texas LLC **Jan. 2025 - Present**

- Review commercial HVAC and plumbing bid drawings; perform material and equipment takeoffs in Bluebeam Revu and Excel and track revisions, scope changes, and bid updates.
- Produced Delta 1 plumbing revisions for Masjid Al-Ekhlis after architectural changes, coordinating fixture and equipment locations, piping, risers, details, schedules, and removal of obsolete work.

Warehouse Associate | Texas Air Products **Jul. 2026 - Present**

- Receive, verify, stage, and load HVAC equipment and components for contractor orders; identify dampers, air-distribution devices, controls, VFDs/starters, curbs, isolators, and accessories.

SELECTED ENGINEERING PROJECTS

Two-Stage Spur-Gear Speed Reducer | Independent Design Project

- Independently designed an 82-hp reducer from 1,200 to 100 rpm (12:1); completed the SolidWorks assembly, AGMA gear checks, shaft fatigue analysis, MATLAB beam-element FEM, and bearing/key selection.
- Calculated a minimum gear-contact fatigue factor of safety of 1.24 and minimum shaft factor of safety of 1.37; checked intermediate-shaft deflection and slope against design requirements.

Arduino DC Motor Speed Control | Team Laboratory Project

- Built and tested a PWM motor-control circuit with potentiometer input, transistor, flyback diode, and LCD; four trials per operating point yielded mean speeds of 743.6-8,225 rpm.

1994 Ford F-150 Restoration | Personal Project

- Restored operation through fuel, ignition, starting, and electrical troubleshooting; replaced tanks, pumps, injectors and starter components and soldered injector-connection wiring.

TECHNICAL SKILLS

Design: SolidWorks parts, assemblies and drawings; AutoCAD; Revit; AGMA analysis; shaft, bearing and key design; Modified Goodman fatigue analysis

Analysis: MATLAB/Simulink; OpenFOAM; ParaView/ParaFoam; Python/Jupyter; Linux; beam-element FEM

Applied tools: Bluebeam Revu; Excel; drawing review; takeoffs; calipers; multimeters; hand and power tools